

VIVA CHEAT SHEET

Impact of UPI-Based Digital Payments on the Velocity of Money and Economic Activity in India

Everything you need to defend the project, in the order you are likely to need it

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How to use this. Learn Parts 1 and 2 until you can say them without looking — between them they cover most of what an examiner actually asks. Part 3 is the general question bank. Part 4 is the five hard questions and deserves the most preparation, because those are the ones an examiner asks when a project is good. Part 5 covers opinion and risk questions, and Part 6 is what to do when you are stuck. Page numbers given as "p.24" refer to the printed pages of the research paper.

Part 1 · The opening answer

Almost every viva starts with some version of "tell me about your project". Have this ready. It is about sixty seconds spoken, and it sets the agenda — examiners usually ask their next question about something you just said, so every sentence here is a door you are choosing to open.

Say this: My project asks whether UPI has changed the speed at which money moves around in India. UPI grew from about 2 crore payments in FY 2016-17 to over 24,000 crore in FY 2025-26. But the size is not the interesting part. The interesting part is that the average payment got smaller, from about ₹1,838 to ₹1,300. That tells me UPI did not just digitise payments that already existed. It added new, very small everyday payments. Using Fisher's equation, more transactions on the same stock of money means higher velocity. I also found that the value settled over UPI rose from about half of India's GDP to about nine-tenths of it in three years. On economic activity I am more careful. 57 per cent of small shopkeepers reported higher sales after taking up digital payment, which points one way. But I cannot prove UPI caused it, because smartphones, internet and incomes all rose in the same decade.

Notice what that answer does. It names a concept (Fisher's identity), gives three specific numbers, states a finding that is genuinely your own, and admits a limit. Examiners reward all four.

The single sentence, if you only get one

Say this: UPI made very small payments worth making. Because there are now far more payments on roughly the same stock of money, money moves around faster.

Part 2 · Numbers to know cold

You will not be asked all of these. You will be asked some of them, and hesitating over a number in your own project looks worse than not knowing a concept. Learn the first block properly; skim the rest.

The essential eight

What	Number	Page
UPI volume, FY 2016-17 → FY 2025-26	2 crore → 24,162 crore	20
UPI value, FY 2016-17 → FY 2025-26	₹0.07 → ₹314 lakh crore	20
Average payment, peak → latest	₹1,838 (FY 2020-21) → ₹1,300	22
UPI value ÷ nominal GDP	0.52× → 0.91× in three years	26
UPI share of retail digital volume	81% (FY 2024-25)	24
India's share of world real-time payments	about 49% (2024)	32
Small merchants reporting higher sales	57%	15
Rural vs urban internet per 100 people	46.73 vs 113.83	34

The supporting set

What	Number
Banks live on UPI	44 (FY 2016-17) → 703 (FY 2025-26)
UPI users onboarded	55.49 crore (June 2026)
Merchants accepting UPI	about 6.5 crore; 56.86 crore QR codes
Composition of UPI volume	P2M is 63%; 86% of P2M is below ₹500
RBI Financial Inclusion Index	64.2 (Mar 2024) → 67.0 (Mar 2025)
RBI Digital Payments Index	465.33 → 493.22 (March 2018 = 100)
PMJDY beneficiaries	55.98 crore; over 55% women
Demonetisation	₹15.4 trillion withdrawn = 86.9% of note value; 98.96% returned
Digital payments around demonetisation	71.27 → 111.45 crore a month (+56%)

What	Number
UPI growth, pandemic years	78% (FY 2020-21), then 106% (FY 2021-22)
Total digital payments	2,071 crore (FY 2017-18) → 18,737 crore (FY 2023-24), CAGR 44%
Cybersecurity incidents	10.29 lakh (2022) → 22.68 lakh (2024)
Users reporting they spend more	about 75% in one survey
MDR on small-merchant UPI	zero up to ₹2,000; Government pays 0.15% incentive

The eight data sets in one line each

If an examiner points at any exhibit and says "explain this", these are your answers. They are the same sentences printed in the call-out boxes in the paper.

Data set	What it shows
1 · p.20	UPI grew about 12,000-fold in volume, and volume grew faster than value every year from FY 2021-22 — so payments were getting smaller.
2 · p.22	The average payment fell from ₹1,838 to ₹1,300 while volume rose ten-fold. This is my most important finding.
3 · p.24	UPI is 81% of retail digital transactions, so retail payment trends in this period can fairly be attributed to it.
4 · p.26	UPI value went from 0.52 to 0.91 times GDP. The level means little; the rise means a lot.
5 · p.28	Demonetisation gave a 56% jump, but 99% of the cash came back — it accelerated adoption without causing it.
6 · p.30	COVID growth was 78% then 106% — faster after the shock than during it, so the habit stuck.
7 · p.32	Financial Inclusion Index 64.2 → 67.0, with usage rising fastest, which is the sub-index that matters.
8 · p.34	Rural internet is 46.73 per 100 against urban 113.83 — the limit is connectivity, not acceptance.

Part 3 · Question bank

A. Topic and motivation

Q. Why did you choose this topic?

Because I use UPI every day, and I wanted to know if something that changed my own habits had changed the economy too. The velocity of money is a concept from our syllabus that nobody can see directly. UPI gave me a way to look at it with real published data.

Q. What exactly is your research question?

There are two questions. First, has UPI changed the speed at which money moves around in India? Second, can I find any effect on real economic activity in published data? I answer the first one more confidently than the second, and I say so in the paper.

Q. What is new in your project? Everyone knows UPI grew.

I agree, the growth is not my finding. My own contribution is the average payment size in Table 2. That figure is not published anywhere. I worked it out by dividing yearly value by yearly volume. It falls from ₹1,838 to ₹1,300, and that fall is what separates "UPI added new payments" from "UPI just digitised old ones".

Q. What period does your study cover, and why?

FY 2016-17 to FY 2025-26. FY 2016-17 was UPI's first full year, and FY 2025-26 was the latest year with complete data when I wrote. Ten years also lets me look at two big shocks, demonetisation and COVID-19.

B. Concepts

Q. What is the velocity of money?

It is the average number of times one rupee is used to buy goods and services in a given period. It is not a physical quality of money. It just shows how quickly people spend what they hold.

Q. State the equation of exchange and explain each term.

$M \times V = P \times T$. M is the money stock, V is velocity, P is the average price level and T is the number of transactions. It is an identity, which means it is true by definition and not an assumption. Its use is that if M stays the same and T goes up, then V must have gone up.

Q. What is the Cambridge approach, and how does it relate to velocity?

It writes $M = k \times P \times Y$, where k is the share of income people keep as money. Velocity is one divided by k . So a rise in velocity and a fall in k are the same statement. This is the version I actually use, because UPI works by letting people keep less cash in hand.

Q. What is UPI, in one sentence?

It is a common set of rules, launched in 2016, that lets any bank account be paid from any app using one short address. It is instant, works all day, and is free for the user.

Q. What is NPCI, and why does it matter that it is not-for-profit?

NPCI is the National Payments Corporation of India. It built UPI and runs it, under RBI guidance. Because it is not-for-profit, it does not have to make a profit on the system itself. That is why UPI is free and card payments are not.

Q. What is interoperability and why was it decisive?

It means a user of one app can pay a user of any other app. Before UPI, wallets were closed loops, so money in one could not reach another. The IMF studied UPI and found that interoperability drove its growth. Users could join through a brand they trusted and switch later, so every company had to keep improving.

Q. Why did cards never achieve what UPI did?

Because of the cost of accepting them. A card machine costs money, and the shopkeeper pays a fee on every sale. That does not work on a ₹20 margin. A printed QR code costs nothing, and below ₹2,000 there is no fee at all.

Q. What is the difference between nominal and real GDP, and which did you use?

Nominal GDP is measured at current prices, so it includes inflation. Real GDP is at constant prices. I used nominal GDP in Table 4, because UPI value is also in current rupees. Both sides of a ratio have to be measured the same way.

C. Method

Q. What kind of data did you use?

Only secondary data. I used published figures from RBI, NPCI, PIB, MoSPI and TRAI, plus research from the IMF, BIS and World Bank.

Q. Why did you not conduct a primary survey?

Because what I am studying is national in size, and it is already measured in full by the bodies that run it. Any survey I could actually do would be too small to stand for the country, and it could not improve on RBI and NPCI data.

Q. How did you make sure your sources were reliable?

I took every number from the body that produced it, not from news or commercial websites. Each figure has a number in brackets pointing to the Bibliography, so anyone can check it.

Q. Which figures are your own calculations?

Two of them. The average value per payment in Table 2, and UPI turnover as a multiple of GDP in Table 4. I have written both methods under the tables so anyone can repeat them.

Q. What are the limitations of your study?

I list seven on page 11. The three that matter most are these. It is all secondary data. Correlation is not causation, because smartphones and incomes rose alongside UPI. And velocity cannot be measured directly, so Table 4 is an approximation and not a measurement.

Q. What were your hypotheses?

Three. H1 says transactions velocity has gone up. H2 says UPI added genuinely new small payments, not just old ones moving across. H3 says the gains are shared unevenly between rural and urban India. All three are supported, and H1 with one qualification.

D. Findings

Q. What is your most important finding?

That the average UPI payment shrank from ₹1,838 to ₹1,300 while volume rose more than ten times. If UPI had only digitised payments that already existed, the average size would have stayed roughly flat. It fell instead. So UPI reached payments it never served before: the tea, the vegetables, the auto fare.

Q. Why does a falling average transaction size matter for velocity?

Because Fisher's equation is about the number of transactions, not their size. Adding a very large number of very small payments raises T a lot while adding little to total value. That is exactly what it means for each rupee to be used more often.

Q. Did demonetisation cause UPI's growth?

It speeded it up, but it did not cause it. Digital payments rose 56 per cent in the seven months after November 2016. But about 98.96 per cent of the notes came back, and cash use recovered. Growth carried on for years afterwards at rates far above 2016-17. Demonetisation removed the effort of trying something new. It did not change the economics.

Q. Why did UPI grow during COVID-19 when the economy shrank?

Two reasons. One is substitution, where payments that would have been cash became digital. The other is new users, where shopkeepers and customers who had resisted learned because there was no choice. New users mattered more, and I can show it. Growth was 78 per cent in FY 2020-21 but 106 per cent in FY 2021-22. If it had been only substitution, it would have reversed once shops reopened. It did not.

Q. Why is demonetisation's effect temporary but COVID's permanent?

It comes down to how long each one lasted. Demonetisation took away the alternative for a few months, so people complied and then went back. The pandemic lasted long enough for digital payment to become a habit, and habits stay after the pressure is removed.

Q. Has UPI increased financial inclusion?

Yes, and we can measure it. The RBI Financial Inclusion Index rose from 64.2 to 67.0 in one year, and the usage part rose fastest. Access was already mostly solved by Jan Dhan, which reached 55.98 crore accounts. The real problem was accounts lying unused, and the World Bank says the share of Indian women with inactive accounts fell from a third in 2021 to 18 per cent in 2024.

Q. Has UPI increased GDP?

I cannot claim that, and I deliberately do not. What I can say is that 57 per cent of small shopkeepers reported higher sales after taking up digital payment. The BIS also finds that fast payment systems raise digital finance use most in lower-income countries. That is evidence pointing towards a real effect. It is not a measurement of one.

Part 4 · The five hard questions

These are the questions an examiner asks when the project is good enough to probe. Each targets a place where the paper is more careful than it strictly needed to be — which is exactly why they are worth marks if you handle them well. Prepare these five above all else.

Q. Hard 1. Which velocity have you measured — income or transactions?

Transactions velocity, and the difference matters. Income velocity counts only payments for newly produced output, so it moves slowly. Transactions velocity counts every payment, including money I send between my own accounts and payments between wholesalers. A payment system affects transactions velocity directly and income velocity only indirectly. Most popular writing mixes the two up, so I labelled mine clearly.

Q. Hard 2. So does 0.91× mean every rupee circulates 0.91 times a year?

No, and that is the trap in the number. The top of the ratio counts every payment. The bottom counts only final output. They are not measuring the same thing, so the level of the ratio means nothing. What does mean something is the change. The mismatch stays about the same each year, so a rise from 0.52 to 0.91 in three years tells me payment activity grew much faster than output.

Q. Hard 3. Some of the rise is just cash payments becoming visible. Doesn't that ruin your finding?

It qualifies my finding, and I say so in the paper. A cash payment that becomes a UPI payment raises the measured figure with no change in behaviour, because cash was never counted. The data cannot separate that from genuinely new payments. But Data Set 2 gives me a second check. If this were only migration, the average payment size would have stayed flat. It fell by 29 per cent. So at least part of the rise is new activity.

Q. Hard 4. Inflation was positive over your period. Doesn't that undermine the falling average?

It actually strengthens it. Rising prices push the average payment up. So inflation works against my finding rather than causing it. The average fell anyway, which means the shift towards small payments is even bigger in real terms.

Q. Hard 5. If you cannot prove causation, what have you actually established?

Three things. First, that payment turnover per rupee of output rose sharply. That is measured, not guessed. Second, that the growth is made of new small payments and not just old ones moving across. That is the average size evidence. Third, that the effect on output is supported but not proved, because smartphones, internet, bank accounts and incomes all grew in the same decade, and published figures cannot separate them. Refusing to claim too much is itself a finding.

If pressed further on Hard 5: To prove cause and effect I would need UPI data by district, and something that varied between them. For example, comparing districts where adoption was forced early with similar ones where it was not. The IMF did something like this and found faster growth where interoperability helped most. That data is not published, which is why one of my suggestions is that it should be released.

Part 5 · Opinion, risks and the awkward ones

Q. Is UPI being free a good thing? Someone must be paying.

The Government pays, through incentive schemes worth about ₹1,500 crore for small merchant payments. Whether that is worth it depends on whether the gains in recorded activity, inclusion and tax collection are bigger than the spending. That is why one of my suggestions is that the scheme should be reviewed publicly from time to time.

Q. What are the risks of a cashless economy?

There are three. Dependence on infrastructure, because a digital payment needs power, a device and a network at the same time, and cash needs none of them. Fraud, with cyber incidents doubling from 10.29 lakh to 22.68 lakh between 2022 and 2024. And exclusion, because a system that needs the internet cannot include more people than the internet does.

Q. Is UPI secure?

The technology is strong. The app is tied to one device, a PIN is needed, and machine learning blocks suspicious payments. The weak point is the user, who can be talked into approving a payment. That is why my suggestion is fraud awareness inside the apps in regional languages, rather than more technical security.

Q. Does UPI make people overspend?

The evidence suggests it can. In one survey about 75 per cent of users said they spent more, and many said digital money felt less real than cash. It is a real cost, especially for low-income families. I treat it as one. The same ease that raises velocity also removes a restraint that cash gave by accident.

Q. Why is rural adoption behind? Is it a lack of QR codes?

No, and that is the point of Data Set 8. Over 5.45 crore payment points have been set up in tier-3 to tier-6 centres. The problem is connectivity. Rural internet is 46.73 subscribers per 100 people against 113.83 in urban areas. So spreading broadband would help more now than paying for more QR codes.

Q. India has 49 per cent of the world's real-time payments. Isn't that just population?

Partly, and I say so. A share of volume says nothing about quality. What makes it notable is the policy difference. Most fast payment systems abroad charge shopkeepers, and India chose not to. So the growth is partly a result of treating payment as a public service.

Q. What would you do differently or study next?

I would want data by district, because national figures hide regional differences completely. If I carried the project further, I would look at whether the payment record is changing credit access for small traders through the Unified Lending Interface. That is where a payment effect would turn into an investment effect.

Q. Did you write this yourself?

Yes. Say it plainly, then show it. Name the two figures you calculated, say why you chose nominal GDP and not real GDP, and explain the base year break in Table 4. Nobody who has not worked with the material can do that.

Part 6 · Traps, and how to get unstuck

Six things not to say

Do not say	Say instead
"UPI increased India's GDP."	"The evidence points that way — 57% of small merchants reported higher sales — but I cannot prove causation."
"Each rupee now circulates 0.91 times."	"The level of that ratio is not meaningful. Its rise from 0.52 to 0.91 is."
"Velocity increased" (unqualified)	"Transactions velocity increased. Income velocity would move far less."
"Demonetisation made India digital."	"It accelerated adoption, but 99% of the cash returned. The lasting change came later."
"Cash is finished."	"Cash is declining as a share of payments, but currency in circulation is still large relative to GDP."
"My data is from the internet."	"My data is from RBI, NPCI, PIB, MoSPI and TRAI publications, each referenced in the Bibliography."

If you do not know the answer

You will be asked something you have not prepared. That is normal and is not a failure. What examiners penalise is bluffing, because it is obvious. Three usable moves:

1. Say what you do know, then mark the limit honestly. "I did not examine that directly. What my data does show is..." — this turns an unknown into an answer.
2. Reason out loud from the identity. Almost every velocity question can be worked from $M \times V = P \times T$. Say what happens to each term and why. Visible reasoning scores better than a remembered fact.
3. Point to the paper. "That is in my limitations on page 11" or "Data Set 8 covers that" is a legitimate answer, and it shows you know your own structure.

The one habit that matters: Whenever you give a number, say where it came from: RBI, NPCI, PIB, TRAI or the IMF. It takes three extra words. It shows the project is really researched, and it makes every answer harder to argue with.

The last thing to read before you go in

Remember: UPI grew from 2 crore to 24,162 crore payments. The average payment fell from ₹1,838 to ₹1,300. Value settled rose from 0.52 to 0.91 times GDP. Small payments got added, so money moves faster. On output, the evidence points one way but I do not overclaim. Everything else is detail.